

Module 7

Design for Printability — Tier 2: Intermediate

1. Overhangs

Any surface angled more than ~45° from vertical without support underneath tends to sag or fail. Design around this by adding gradual angles or chamfers instead of sharp horizontal overhangs.

2. Tolerances

FDM printing isn't perfectly precise — parts that need to fit together (like a lid and a box) need small gaps built in.

- General rule: 0.2–0.3mm clearance for a snug fit
- 0.4mm+ clearance for an easy, loose fit

3. Orientation

How you orient a model on the bed affects strength, surface finish, and whether supports are needed at all.

- Parts are weakest between layers (Z-axis) — orient to keep stress along the layers, not across them
- Flat-bottom orientation avoids supports where possible
- Vertical orientation gives better surface detail on curved surfaces

4. Wall Thickness

Design walls at least 2x your nozzle diameter (commonly 0.8mm minimum for a 0.4mm nozzle) — thinner walls may not print reliably or at all.

End of Module 7 — Next: Module 8, Advanced Slicer Settings